

Introduction to Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with large language models. Instead of relying only on pre-trained knowledge, RAG systems retrieve relevant documents from a database and use them as context before generating answers.

A typical RAG system has three main components:

1. Document Loader
2. Embedding Model
3. Vector Database

The embedding model converts text into numerical vectors. These vectors are stored in a vector database such as FAISS or Pinecone. When a user asks a question, the system converts the question into a vector and retrieves the most similar documents.

The retrieved documents are then passed to a large language model like Gemini or GPT to generate a final answer. This approach improves accuracy and reduces hallucinations.

Advantages of RAG:

- Uses private knowledge securely
- Reduces hallucination
- Provides context-aware answers
- Can be updated without retraining the model